

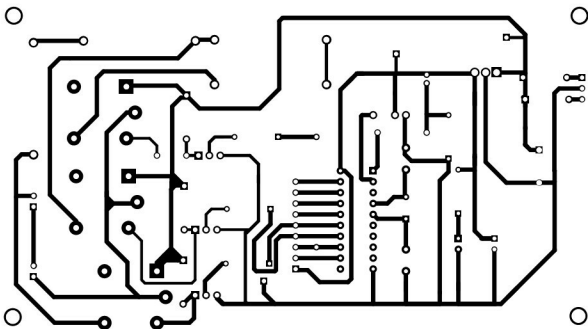


Fig. 3: Actual-size PCB pattern of the remote-controlled smartfan using AT89C2051

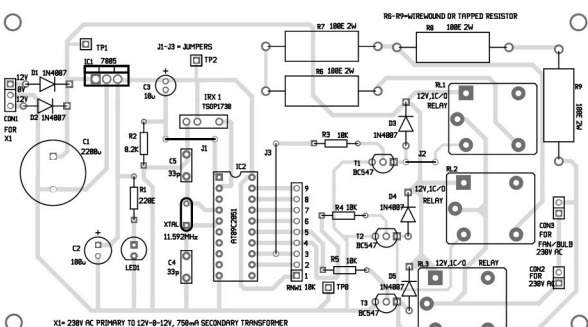


Fig. 4: Component layout of the PCB

and on-chip analogue comparator. 12V electromagnetic relays act as switching regulators for positions tapped resistors across R6 through R9 to drop voltage levels to control fan or bulb. TSOP1738 is a part of IR remote control receiver, which receives IR remote signals and gives IR pulses to AT89C2051 MCU. Output of TSOP1738 is active low,

and it gives +5V in off state.

The mains supply is stepped down by transformer X1, rectified by a full-wave rectifier comprising D1 and D2, filtered by C1 and



Fig. 5: Wire-wound resistor (tapped resistor) prototype

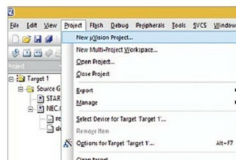


Fig. 6: Create Project window